

IM1 Book 1 Selected Answers

IM1 Dream Team

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1. (a) $A_{ABCD} = 25$, $A_{BCEF} = 9$
(b) –
(c) –
(d) $A = 34$
(e) $l = \sqrt{34}$
(f) –
2. $l = 4\sqrt{5}$
3. –
4. $AB = \sqrt{41}$
5. $l = 5\sqrt{2}$
6. $l = \sqrt{5}$, No
7. $d = 10\sqrt{2}$
8. 12
9. $(12, 2), (2, 2)$
10. No
11. (a)

$$\begin{aligned}SR &= PQ = 13 \\RQ &= PS = 17 \\m_{SR} &= m_{PQ} = \frac{-5}{12} \\m_{RQ} &= m_{PS} = \frac{15}{8}\end{aligned}$$

- (b) It is a parallelogram but not a rectangle.
12. $(-6, 3), (10, 3)$

13. $AB = BC = \sqrt{10}$
14. (a) $(5, 0)$ – Answers may vary.
 (b) $(5, 1)$ – Answers may vary.
 (c) $x = 5$
 (d) –
15. (a) $AP = BP = 2\sqrt{5}$
 (b) $(2, 2), (3, 5), (4, 8)$
 (c) No
 (d) $y = 3(x - 1) - 1$
16. (a) $(2, 5), (3, 4)$ – Answers may vary.
 (b) $(1, 4), (2, 3)$ – Answers may vary.
17. $(1, 5)$ – Answers may vary.
 There are infinite options.
18. (a) Same size, different shape, incomparable orientations – the pieces aren't equal.
 (b) Same size, same shape, different orientations – the pieces are equal

Definition: Two figures are congruent if they have the same size and shape.

19. $y = \frac{1}{3}(x - 2) + 2$
20. $(0, 5), (3, 4), (4, 3), (5, 0), (4, -3), (3, -4), (0, -5), (-3, -4), (-4, -3), (-5, 0), (-4, 3), (-3, 4)$
21. $(2, 3), (2, -3)$
22. (a) Answers may vary: $(0, 0), (2, 5)$.
 (b) No
23. Answers may vary: $(0, 0), (4, 1), (8, 0), (4, -1)$
24. $(0, 5 + 4\sqrt{2}), (0, 5 - 4\sqrt{2})$
25. 6
26. Leg 1: $24 - 12\sqrt{2}$
 Leg 2: $24 - 12\sqrt{2}$
 Hypotenuse: $24\sqrt{2} - 24$
27. Yes
28. (a) If point P is equidistant from the coordinate axes, then point P is on the line $y = x$.

- (b) No
 - (c) Answers may vary. If quadrilateral $ABCD$ is a square, then quadrilateral $ABCD$ is a rectangle.
 - (d) Answers may vary. If quadrilateral $ABCD$ has four right angles, then quadrilateral $ABCD$ is a rectangle.
29. They have the same side lengths.
30. (a) $\triangle ABC$ must be the same size and shape as $\triangle DEF$.
If two triangles have the same three side lengths, then the two triangles are congruent.
- (b) $\overline{KL}$
- (c) Typo: The question should say “What angle of $\triangle ABC$ corresponds to $\angle LKM$?”
Answer: $\angle BAC$
- (d) $\overline{GI} \cong \overline{GI}$, so there are three pairs of congruent segments.
- (e) They are congruent.
31. –
32. $\angle ABC$
33. $\overline{AB}$
34. (a) $PNMRQ$
(b) $\angle Q$
35. They must add to 90° .
36. It must be 90° .
37. $d = \sqrt{(x-2)^2 + (y-4)^2}$
38. (a) –
(b) –
(c) –
(d)

$$m_{QR} = \frac{-a}{b}$$

$$m_{QP} = \frac{b}{a}$$

39. –

40. –

41. –

42. –

43. –

44.

$$\triangle LAG \cong \triangle LEB$$

$$\triangle LEG \cong \triangle LAB$$

$$\triangle AGB \cong \triangle EBG$$

$$\triangle AGE \cong \triangle EBA$$

45. –

46. (a) $y - 2 = \frac{1}{2}(x + 1)$

(b) $y = \frac{4}{5}(x - 2)$

(c) $y = -\frac{5}{4}(x - 2)$

47. (a) –

(b) Q; C.P.C.T.C.

48. $n = \frac{49}{4}$

49. (a) $\overline{AC} = \overline{AB} + \overline{BC}$; segment addition

(b) –

50. Yes

51. No

52. C.P.C.T.C.

53. (a) $\angle ABD + \angle DBC = \angle ABC$; angle addition

(b) –

56. $m = -1$

57. –

58. –

59. (a) $\sqrt{(x+1)^2 + (y-5)^2}$

(b) $\sqrt{(x-5)^2 + (y-2)^2}$

(c) $\sqrt{(x+1)^2 + (y-5)^2} = \sqrt{(x-5)^2 + (y-2)^2}$

(d) $y = 2x - \frac{1}{2}$

(e) $(2, \frac{7}{2})$

(f) $m_{AB} = -\frac{1}{2}$

(g) $-$

60. All points (x, y) that are equidistant to $(3, 5)$ and $(7, -1)$.

61. (a) $y - 5 = \frac{5}{4}(x - \frac{1}{2})$

(b) $(\frac{9}{2}, 10)$

(c) $\overline{PA} = \overline{PB}$

62. $-$

63. $(-\frac{7}{2}, -\frac{1}{2})$